MILITARY SPECIFICATION

TRANSISTOR, NPN, SILICON

TYPE 2N1072

1. SCOPE

1.1 This specification covers the detail requirements for diffused silicon, NPN, junction transistors with the following salient characteristics at an ambient temperature of (25 ± 3)°C (see 3.2 of MIL-S-19500):

	h_{FE}	$V_{CE}(sat)$	$V_{BE}(sat)$	$t_d + t_r$	$t_s + t_f$
	V_{CE} = 5 Vdc	I_C = 750 mAdc	I_C = 750 mAdc	Method A	R_C = 25 ohms
	I_C = 750 mAdc	I_B = 75 mAdc	I_B = 75 mAdc	V_{CC} = 20 Vdc	$I_{B(1)}$ = 75 mAdc
				$V_{BE(0)}$ = 0	$I_{B(2)}$ = -75 mAdc
Min.	20	0.50 Vdc	0.70 Vdc	-	-
Max.	-	2.0 Vdc	1.8 Vdc	100 μsec	300 μsec

1.2 Absolute Maximum Ratings

P_C 1/	P_C 2/	T_{stg}	T_J	V_{CES} 3/	V_{CB} 3/	I_E	V_{EB} 3/
12.5 W	2.0 W	-65°C to +150°C	+150°C	75 Vdc	75 Vdc	2.0 A	6.0 Vdc

1/ This collector power dissipation is for an expected life > 1000 hours with the transistor firmly clamped to a heat sink maintained at 25 ± 3°C. For collector power ratings at higher temperatures P_C is derated at 1 watt for each 10°C rise in heat sink temperature.

2/ For unit in free air at 25°C with no heat sink, derate at 1 watt for each 65°C increase in ambient.

3/ These voltages may be exceeded (without permanently impairing transistor performance) provided current is limited to 10.0 mAdc.

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2. APPLICABLE DOCUMENTS

2.1 The following military specification and its ancillary documents, of the issue in effect on the date of invitation for bids, forms a part of this specification:

MIL-S-19500 - Semiconductor Devices, General Specification for.

(Copies of Specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 General requirements. Requirements shall be in accordance with specification MIL-S-19500 and as specified herein.

3.2 Abbreviation and symbols. The abbreviations and symbols used herein are defined in specification MIL-S-19500.

3.3 Design and Construction. Transistors shall be of the design, construction and physical dimensions specified in figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in Tables I, II and III herein.

3.5 Marking. Transistors shall be marked with the manufacturer's identification, "USA" prefix, type designation, acceptance date, and lot identification suffix letter in accordance with MIL-S-19500.

3.5.1 Lot Identification Suffix Letter. The lot identification suffix letter shall appear on each transistor immediately following the acceptance date. A suitable capital Gothic letter shall be chosen by the manufacturer such that each acceptance inspection lot is uniquely identified by the acceptance date and lot identification suffix letter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection conditions. Unless otherwise specified, electrical measurements shall be taken within 3 seconds of application of bias.

4.2 Qualification inspection. Qualification inspection will be performed at a laboratory designated by the Government, and shall consist of the examinations and tests specified in Tables I, II and III.

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4.3 Acceptance inspection. Acceptance inspection shall consist of groups A, B and C, as specified in Tables I, II and III respectively, and inspection or preparation for delivery. (see 4.3.7).

4.3.1 Sample. The minimum sample size used to represent a lot for groups A and B inspection shall be that which is required for a defect acceptance number of 1 and a rejection number of 2. For groups B and C inspection, a different set of units shall be selected for each subgroup.

4.3.2 Acceptance-inspection information. One copy of the acceptance-inspection information pertinent to the transistor inspection lot shall be furnished by the transistor supplier and shall accompany each transistor shipment from the inspection lot to the equipment manufacturer.

4.3.3 Supplier's quality-assurance procedure. At the discretion of the Government, the supplier may be required to grant access to the Government, to permit thorough evaluation of the supplier's quality-assurance procedures with regard to transistor inspection lots intended for delivery on the following:

(a) Direct purchases by or direct shipments to the Government.

(b) Contracts or orders between the transistor supplier and equipment manufacturers.

4.3.4 Action in case of lot rejection. When an inspection lot is rejected on groups B and C inspection, the supplier shall immediately investigate the cause of failure and take corrective action, if necessary, to assure that subsequent lots do not contain excessive defects.

4.3.5 Life test failure after shipment. In the event a sample fails the operation-life or storage-life test, and the inspection lot represented has been shipped prior to the completion of the operation-life or storage-life test, the supplier shall immediately notify the Government of the failure and the action taken.

4.3.6 Disposition of defective transistors. Transistors which are known to be defective shall not be delivered on the contract or order. Lots of transistors which are known to contain an excessive number of defectives shall not be mixed with other lots prior to acceptance inspection.

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4.3.7 Inspection of preparation for delivery. Sample items and packs shall be selected and inspected in accordance with specification MIL-P-19491 to verify conformance with requirements with section 5 of this specification.

4.3.8 Destructive tests. The lead fatigue test of group B inspection and salt atmosphere test of group C inspection are considered to be destructive. The tests of subgroup 2 + 3 of group B inspection can be considered non-destructive if sufficient evidence is presented to the inspection authority to that effect. Acceptable evidence, for example, would be the repeating of all subgroup 2 and 3 tests ten times without significant device degradation.

5. PREPARATION FOR DELIVERY

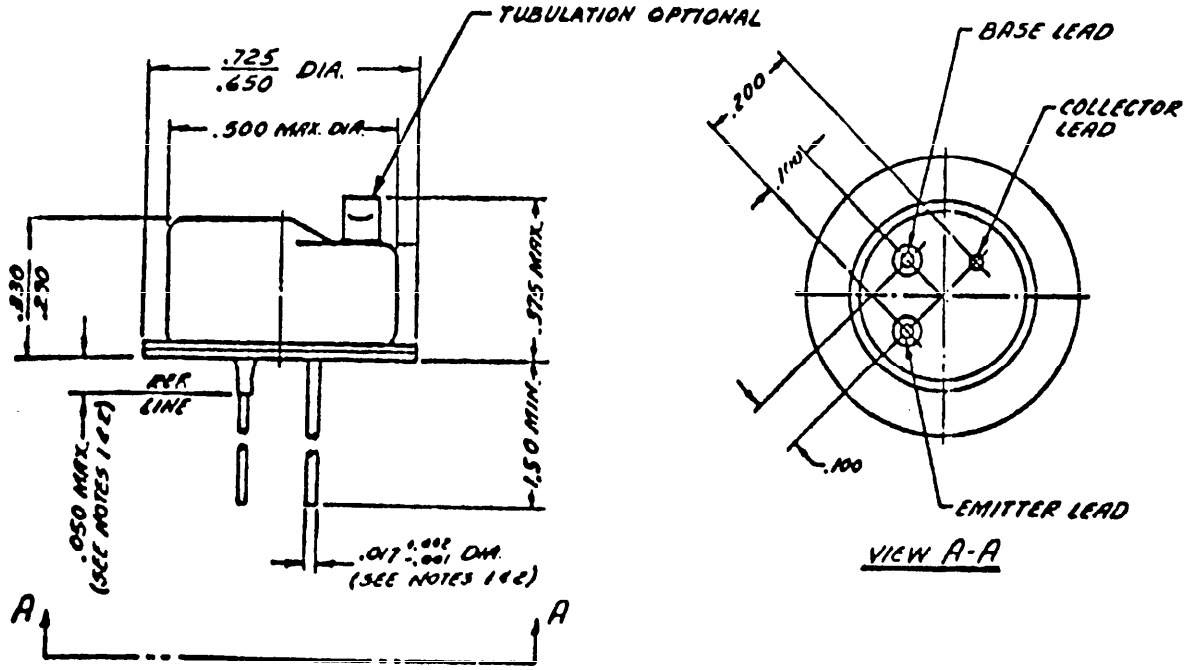
5.1 Unless otherwise specified in the procurement document, preservation, packaging, packing and marking shall be by level C of MIL-P-19491.

6. NOTES

6.1 Disposition of defective transistors. Meeting the sample acceptance requirements is not to be interpreted as permission to deliver transistors which are known to be defective.

NOTICE: When the Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the use, or sell any patented invention that may in any way be related thereto.

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NOTES -

1 - AT MAXIMUM MATERIAL CONDITION LEADS SHALL BE WITHIN .010 OF THEIR TRUE CENTERS WITH RESPECT TO THE .725 DIAMETER MEASURED AT .050 MAX. BELOW SEAT.

2 - THE SPECIFIED LEAD DIAMETER APPLIES IN THE ZONE BETWEEN .050 AND .250 FROM THE SEAT. THE REMAINDER OF THE LEAD SHALL NOT EXCEED .021 MAX. DIAMETERS OF LEADS ABOVE REFERENCE LINE ARE NOT RESTRICTED.

3 - ALL DIMENSIONS IN INCHES

FIGURE 1 - TRANSISTOR, TYPE 2N1072

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TABLE I - GROUP A INSPECTION

EXAMINATION OR TEST	CONDITIONS		SYMBOL	LIMITS		UNITS
	Ref. Para. MIL-S-19500	Specific Conditions		Min.	Max.	
Subgroup 1						
Visual and mechanical examination	C30.13	-	-	-	-	-
Subgroup 2						
Collector to emitter breakdown voltage, short circuit	C50.3	I_C = 1.0 mAdc	BV_{CES}	75	-	Vdc
Emitter to base breakdown voltage, open collector	C50.1	I_E = 100 µAdc	BV_{EBO}	6.0	-	Vdc
Collector cutoff current, open emitter	C50.6	V_{CB} = 5 Vdc	I_{CBO}	-	0.1	µAdc
Static forward current transfer ratio	C50.40	I_C = 750 mAdc V_{CE} = 5 Vdc	h_{FE}	20	-	-
Output capacitance	C50.19	I_E = 0 V_{CB} = 20 Vdc f = 10 kc	C_{ob}	30	55	µµf
Subgroup 3						
Saturation voltage (collector to emitter)	C50.25	I_C = 750 mAdc I_B = 75 mAdc	$V_{CE}(sat)$	0.50	2.0	Vdc

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TABLE I - GROUP A INSPECTION (CONT'D)

EXAMINATION OR TEST	CONDITIONS		SYMBOL	LIMITS		UNITS
	Ref. Para. MIL-S-19500	Specific Conditions		Min.	Max.	
Subgroup 3 (Cont'd)						
Saturation voltage (Base to emitter)	C50.25	I_C = 750mAdc I_B = 75 mAdc	V_{BE}(sat)	0.70	1.8	Vdc
Delay and rise time	C50.21.1 C50.21	Method A R_C = 25 ohms V_{CC} = 20 Vdc V_{BE} (0) = 0 $I_{B(1)}$ = 75 mAdc	$t_d + t_r$	-	100	msec
Storage and fall time	C50.21.1 C50.21	Method A R_C = 25 ohms V_{CC} = 20 Vdc $I_{B(1)}$ = 75 mAdc $I_{B(2)}$ = -75 mAdc	$t_s + t_f$	-	300	msec

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TABLE II - GROUP B INSPECTION

EXAMINATION OR TEST	CONDITIONS		SYMBOL	LIMITS		UNITS
	Ref. Para. MIL-S-19500	Specific Conditions		Min.	Max.	
Subgroup 1						
Physical dimensions	C30.9	-	-	-	-	-
Subgroup 2						
Soldering	C40.13	-	-	-	-	-
Temperature cycling	C40.14	5 cycles $T_{(high)}$ = 150° ±5°C	-	-	-	-
Thermal shock (glass strain)	C40.16	$T_{(high)}$ = 85° $^{+5°}_{-0°}$C $T_{(low)}$ = 0° ± 2°C	-	-	-	-
Moisture resistance	C40.6	-	-	-	-	-
End Points:						
Saturation voltage (Collector to emitter)	C50.25	I_C = 750 mAdc I_B = 75 mAdc	$V_{CE}(sat)$	-	2.2	Vdc
Static forward current transfer ratio	C50.40	I_C = 750 mAdc V_{CE} = 5 Vdc	h_{FE}	15	-	-
Collector to emitter breakdown voltage, short circuit	C50.3	I_C = 2.0 mAdc	BV_{CES}	75	-	Vdc
Subgroup 3						
Shock	C40.10	5 blows X_1, Y_1, Y_2 & Z_1	-	-	-	-
Constant acceleration	C40.4	15,000 G	-	-	-	-
Vibration fatigue	C40.18	V_{CB} = 70 Vdc	-	-	-	-

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TABLE II - GROUP B INSPECTION (CONT'D)

EXAMINATION OR TEST	CONDITIONS		SYMBOL	LIMITS		UNITS
	Ref. Para. MIL-S-19500 Appendix C	Specific Conditions		Min.	Max.	
Subgroup 3 (Cont'd)						
Vibration, variable frequency	C40.20	-	-	-	-	-
End Points: (Same as for subgroup 2)						
Subgroup 4						
Lead fatigue	C40.5	8 ± 0.5 oz. No lead restrictions	-	-	-	-
Subgroup 5 1/						
Thermal resistance	C30.11	-	θ_{J-C}	-	10	°C/W
Burn out		I_R = 2.0 Adc I_C = 0	V_{BE}	1.0	3.0	Vdc
Subgroup 6						
Storage life	C40.7 C40.7.2 C40.7.4	Method B λ = 10% T_A = 150 $^{+0}_{-5}$°C	-	-	-	-
End Points: (Same as for subgroup 2)						

1/ Burn out test shall be performed with the unit clamped to a heat sink maintained at 25°C. A current of 2.0 amperes d.c. shall be passed from base to emitter (in the forward direction of the emitter junction) for a period of 1 minute. At no time during this period shall V_{BE} exceed the specified limits.

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TABLE III - GROUP C INSPECTION

EXAMINATION OR TEST	CONDITIONS		SYMBOL	LIMITS		UNITS
	Ref. Para. MIL-S-19500	Specific Conditions		Min.	Max.	
Subgroup 1		period = 12 mos. 4% AQL				
Saturation voltage (collector to emitter)	C50.25	I_C = 750 mAdc I_B = 75 mAdc T_J = 150°C	V_{CE} (sat)	-	3.0	Vdc
Subgroup 2		period = 12 mos. 4% AQL				
Salt Atmosphere (corrosion)	C40.8	-	-	-	-	-
Subgroup 3	C40.7 C40.7.2	Method B λ = 10% period = 6 mos.				
Operation life.	C40.7.3	T_A = 100°C	-	-	-	-
End Points: (same as for subgroup 2 of Table II)		V_{CC} = 20 ± 2V R_L 25 ± 2.5 ohm I_B (on) = 100 ± 25 MA V_B (off) = 3^{+3}_{-2} V t (on) = 8^{+0}_{-3} msec t (off) = 8^{-0}_{+3} msec				

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